American Computer Science League

2025-2026 ● Contest 1: Short Problem Solutions ● Elementary Division

1. Computer Number Systems To convert from octal to hexadecimal, it is easiest to use 3 bits for octal and then 4 bits for hex. $7654_8 = 111\ 110\ 101\ 100_2 = 1111\ 1010\ 1100_2 = FAC_{16}$	D. FAC
2. Computer Number Systems To convert from base 16 to base 10, multiply each digit by a power of 16. $1 * 16^2 + 2 * 16 + 5 = 256 + 32 + 5 = 293$	C. 293
3. Computer Number Systems To convert 2025_{10} from base 10 to base 8, divide by 8 and use the remainders for the new number. 2025/ 8 = 253 r 1, 253 / 8 = 31 r 5, 31 / 8 = 3 r 7, 3 / 8 = 0 r 3 Reading backwards, the answer is 3751.	B. 3751
4. Computer Number Systems Counting in base 8 from 2025 to 2050 yields 2025, 2026, 2027, 2030, … 2037, 2040, …, 2047, 2050. There are 23 (3 in 2025-2027, 9 in each of 2030-2037 and 2040-2047, and 2 in 2050) of them.	E. 23
5. Computer Number Systems 58 and 64 are the green components. In base 10, 5*16+8=88 and 6*16+4=100 so (88+100)/2=94. Without doing any math, counting from 58 to 64 in hex is 58, 59, 5A, 5B, 5C, 5D, 5E, 5F, 60, 61, 62, 63, 64. The one in the middle is $5E_{16} = 5 * 16 + 14 = 94_{10}$.	C. 94
6. Computer Number Systems Subtracting A from each component in base 16 yields B9-A=AF, 78-A=6E, and 4F-A=45 for the hexadecimal value of $AF6E45_{16}$.	A. AF6E45